

Addis Ababa University

Advanced programming for Bioinformatics

Assignment Module 6

Part I: Conceptual Questions

Question 1

Define the following algorithmic design paradigms and give one real-world example for each.

- a) Brute Force
- b) Divide and Conquer
- c) Greedy Algorithm
- d) Dynamic Programming
- e) Backtracking

Question 2

Differentiate between:

- Greedy Algorithms and Dynamic Programming
- Backtracking and Branch & Bound
- Deterministic and Randomized Algorithms

Use a comparison table.

Question 3

Suppose you are solving the following problems. Which algorithmic paradigm would you choose? Justify your answer.

Problem	Paradigm
Sorting 10 million integers	
Finding shortest path between cities	
Protein sequence alignment	
Genome assembly	
DNA motif discovery	
Large phylogenetic tree estimation	

Part II: Programming (40 Marks)

Implement the following algorithms in Python.

Question 4

Implement Merge Sort using Divide and Conquer.

Requirements

- Input a random list of integers.
- Display the original array.
- Display the sorted array.
- Count the number of recursive calls.

Part III: Bioinformatics Applications

Question 5

Write a Python program that searches for a DNA motif using a brute-force approach.

Input

DNA

ATCGTAGCTAGCTAGCTA

Motif

TAG

Output

- Position(s) where the motif occurs
- Number of comparisons made

Then explain why BLAST is faster than this brute-force approach.

Question 6

A researcher wants to identify a common motif among the following DNA sequences.

ATCGTACGATGCA

GTCGTACCATGGA

TTCGTACGGTACA

ATCGTACTATGCG

Tasks

1. Explain how a backtracking algorithm searches for the motif.
2. Draw the search tree.
3. Explain where pruning occurs.
4. Compare the approach with brute force.